

Chapter 55

Ecosystems

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

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Overview: Ecosystems

- An **ecosystem** consists of **all the organisms living in a community**, as well as **the abiotic factors with which they interact**.
- Ecosystems range from a microcosm, such as an aquarium, to a large area such as a lake or forest.
- Regardless of an ecosystem's size, its dynamics involve two main processes: **energy flow** and **chemical cycling**.
- *Energy flows through ecosystems while matter cycles within them.*

Physical laws govern energy flow and chemical cycling in ecosystems

- Laws of physics and chemistry apply to the **transformations of energy and matter** within the ecosystems, particularly **energy flow**.
- The **first law of thermodynamics** states that energy cannot be created or destroyed, only transformed. *Energy enters an ecosystem as solar radiation, is conserved, and is lost from organisms as heat.*
- The **second law of thermodynamics** states that every exchange of energy increases the **entropy** of the universe. In an ecosystem, energy conversions are not completely efficient, and some energy is always lost as heat.

Conservation of Mass

- The **law of conservation of mass** states that matter cannot be created or destroyed.
- Chemical elements are continually recycled within ecosystems.
- In a forest ecosystem, most nutrients enter as dust or solutes in rain and are carried away in water.
- *Ecosystems are open systems, absorbing energy and mass and releasing heat and waste products.*

Energy, Mass, and Trophic Levels

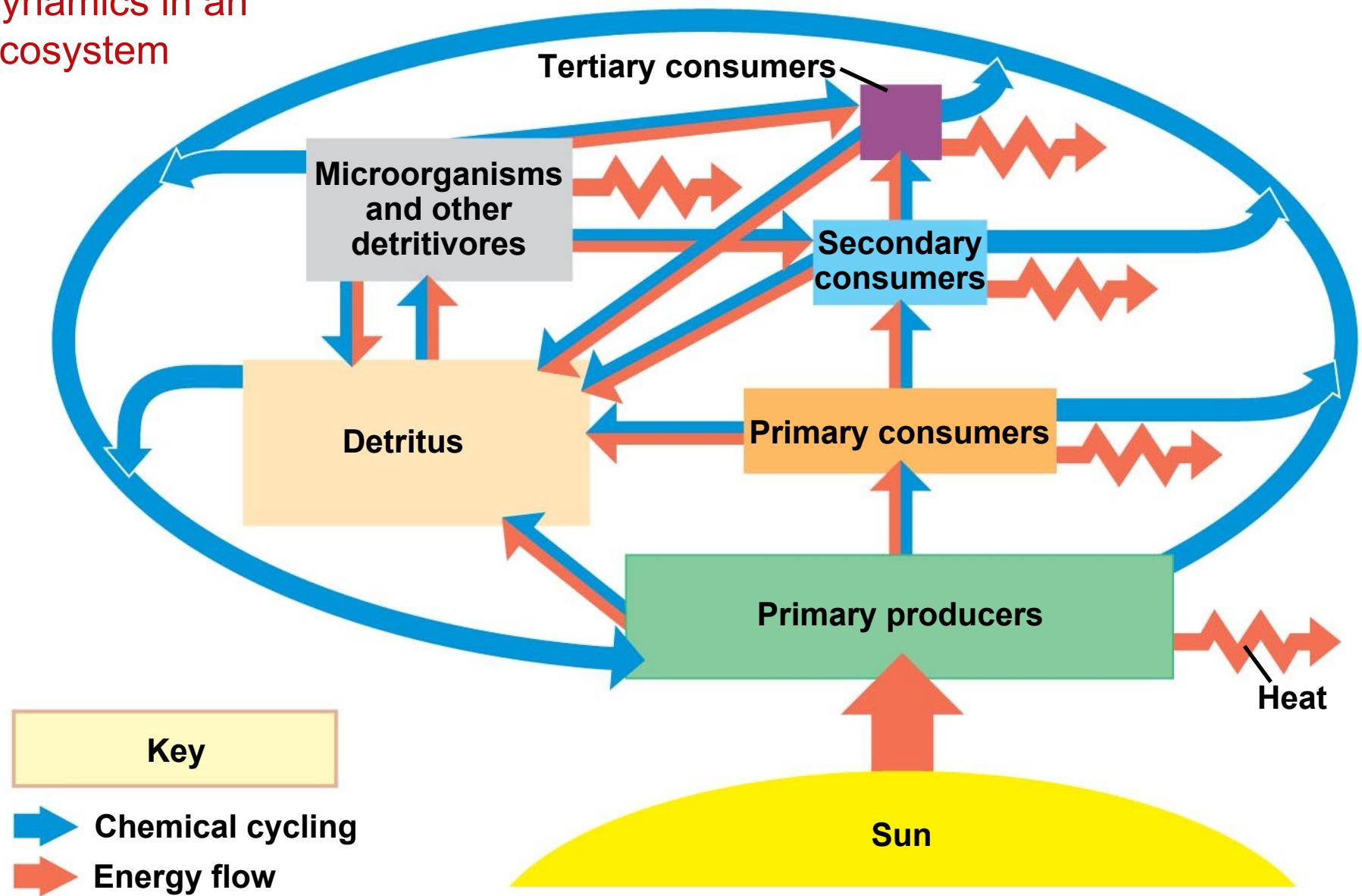
- **Autotrophs** build organic molecules themselves using photosynthesis or chemosynthesis as an energy source.
- **Heterotrophs** depend on the biosynthetic output of other organisms.
- Energy and nutrients pass from **primary producers** (autotrophs) to **primary consumers** (herbivores) to **secondary consumers** (carnivores) to **tertiary consumers** (carnivores that feed on other carnivores).

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- **Detritivores**, or **decomposers**, are consumers that derive their energy from **detritus**, nonliving organic matter.
 - Prokaryotes and fungi are important detritivores.
 - Decomposition connects all trophic levels.

Fungi decomposing a dead tree



Energy and Nutrient Dynamics in an Ecosystem



Energy and other limiting factors control primary production in ecosystems

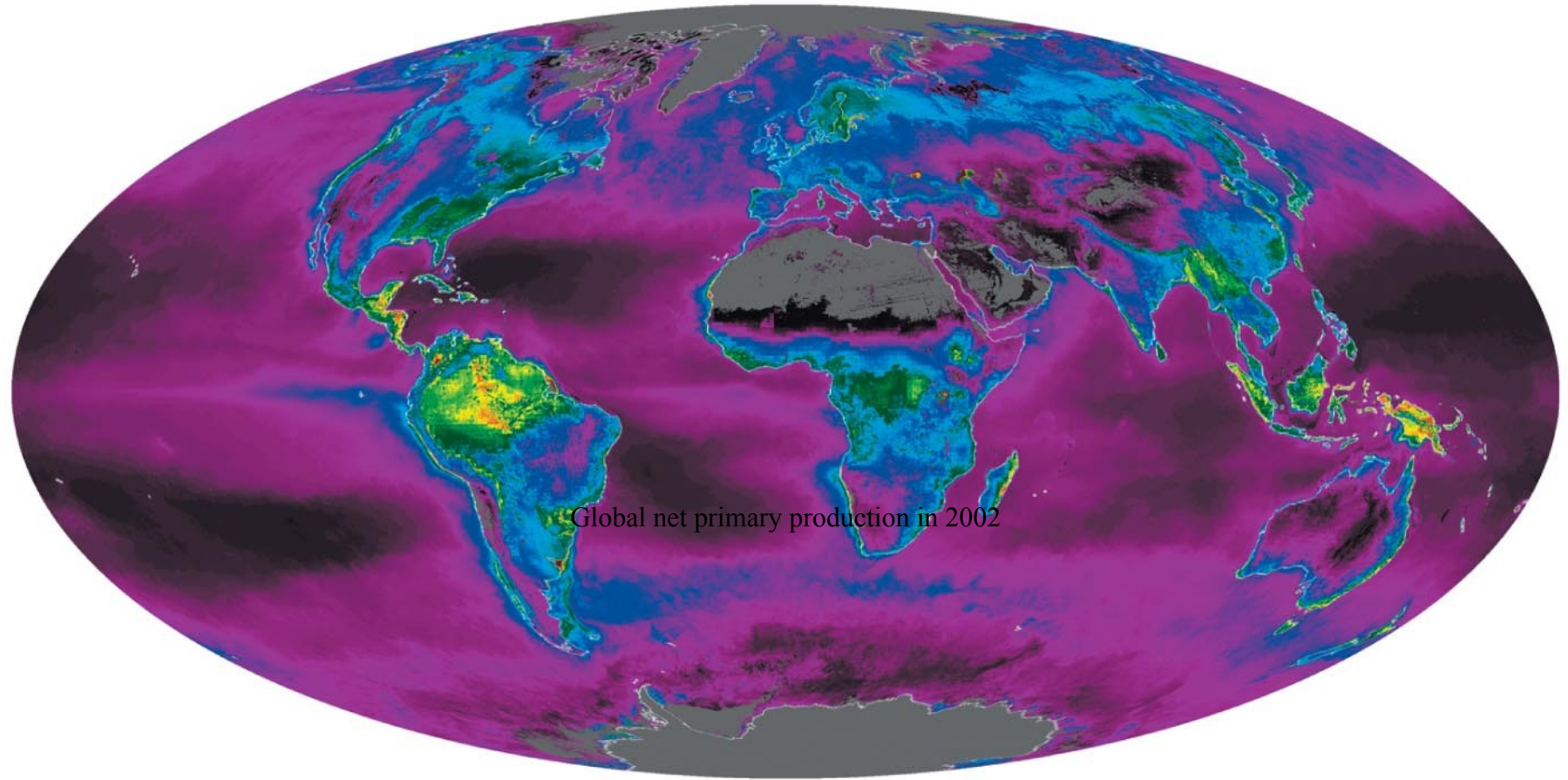
- **Primary production** in an ecosystem is the amount of light energy converted to chemical energy by autotrophs during a given time period. The extent of **photosynthetic production** sets the spending limit for an **ecosystem's energy budget**.
- The amount of solar radiation reaching the Earth's surface limits photosynthetic output of ecosystems.
- Only a small fraction of solar energy actually strikes photosynthetic organisms, and even less is of a usable wavelength.

Gross and Net Primary Production

- Total primary production is known as the ecosystem's **gross primary production = GPP**
- **Net primary production = NPP** is GPP minus energy used by primary producers for respiration.
NPP = GPP - Respiration
- Only NPP is available to consumers.
- ***Standing crop*** is the total biomass of photosynthetic autotrophs at a given time.

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- Tropical rain forests, estuaries, and coral reefs are among the most productive ecosystems per unit area.
 - Marine ecosystems are relatively unproductive per unit area, but contribute much to global net primary production because of their volume.

Global net primary production in 2002



Net primary production (kg carbon/m²·yr)



0

1

2

3

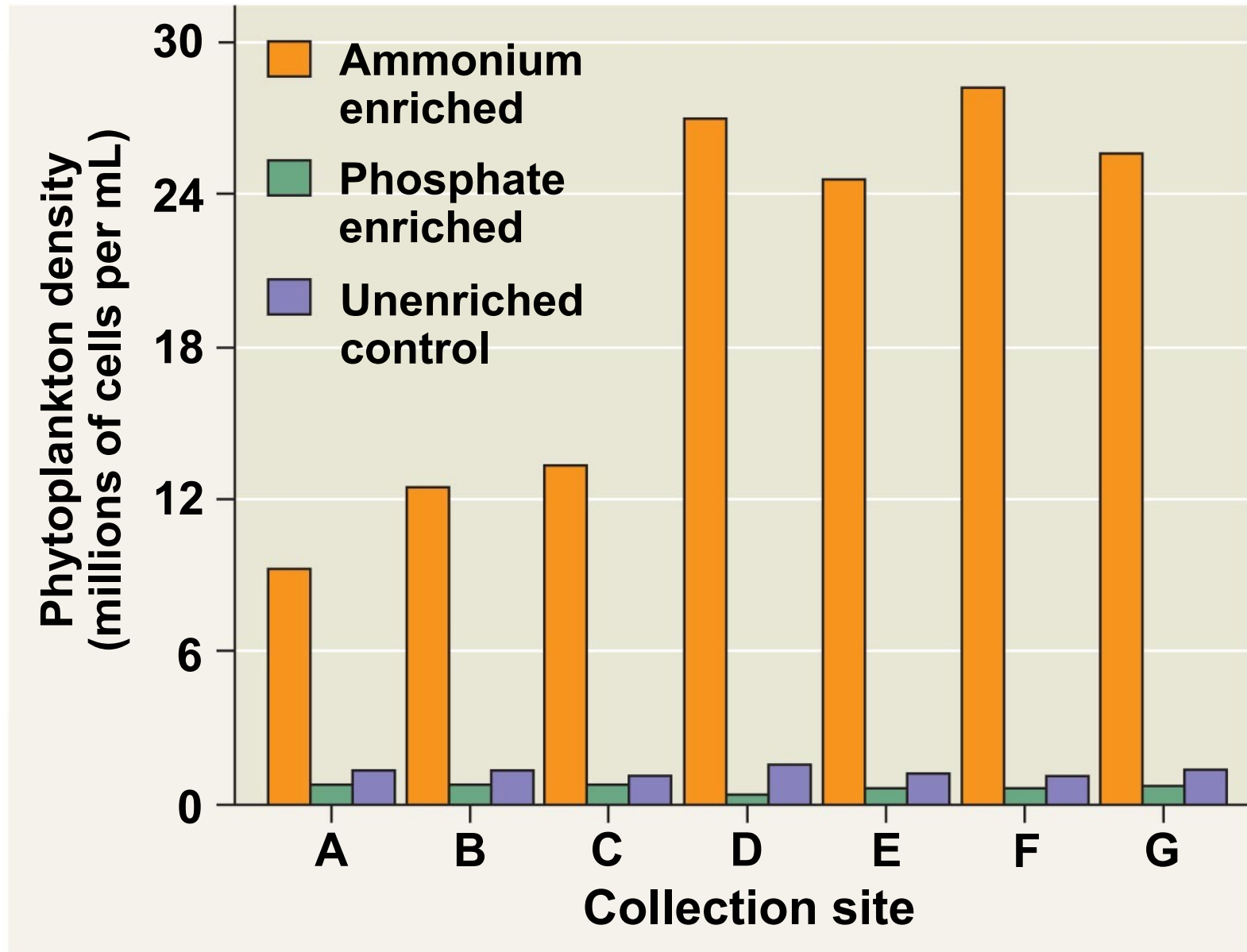
Primary Production in Aquatic Ecosystems

- In marine and freshwater ecosystems, both light and nutrients control primary production.
- Depth of light penetration affects primary production in the photic zone of an ocean or lake.

Nutrient Limitation

- More than light, nutrients limit primary production in geographic regions of the ocean and in lakes.
- **A limiting nutrient** is the element that must be added for production to increase in an area.
- ***Nitrogen*** and ***phosphorous*** are typically the ***nutrients*** that most often ***limit marine production***.
- Nutrient enrichment experiments confirmed that nitrogen was limiting phytoplankton growth.

Which nutrient limits phytoplankton production along the coast of Long Island?

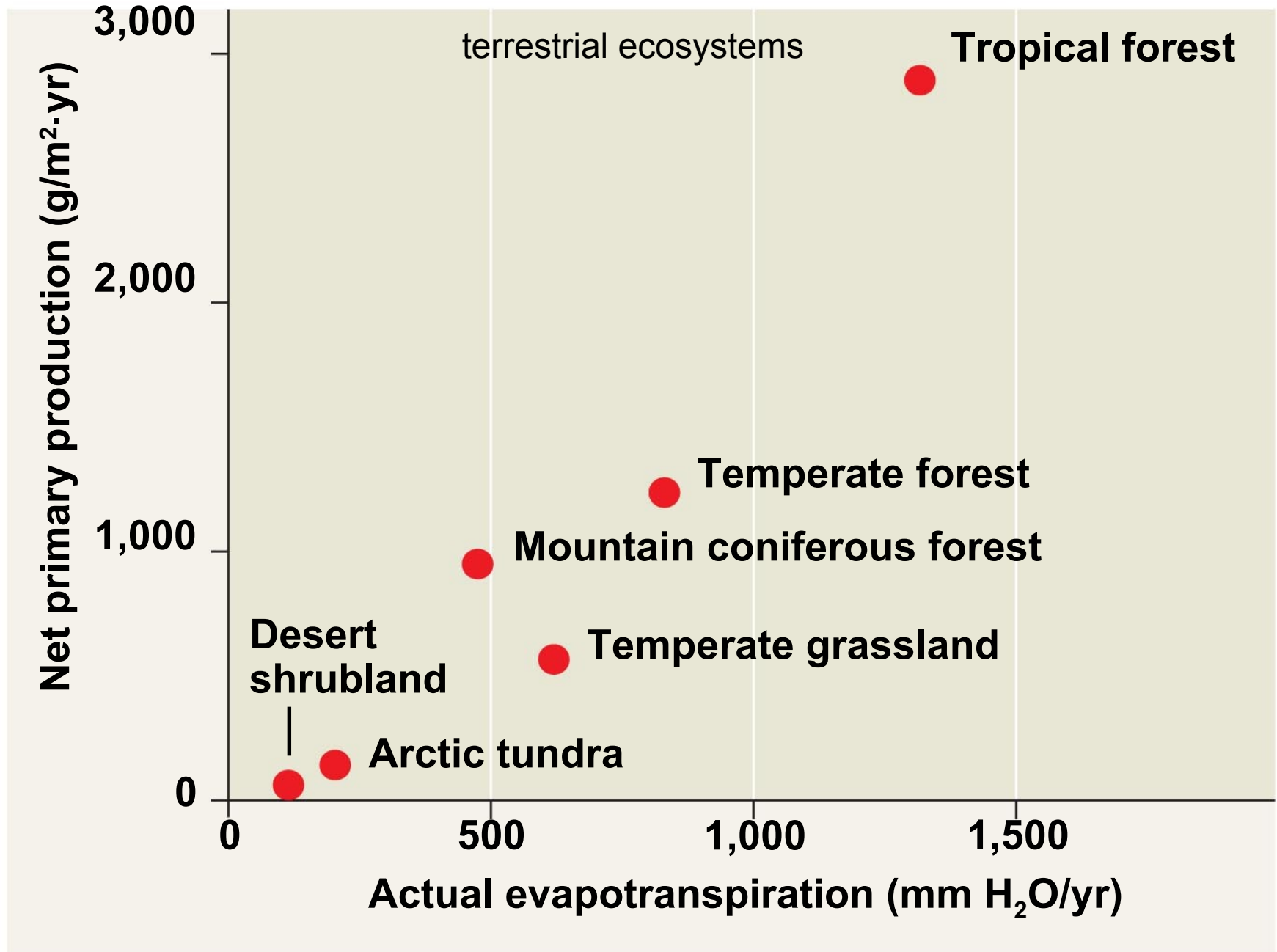


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- Upwelling of nutrient-rich waters in parts of the oceans contributes to regions of high primary production.
 - The addition of large amounts of nutrients to lakes has a wide range of ecological impacts.
 - In some areas, sewage runoff has caused eutrophication of lakes, which can lead to loss of most fish species.

Primary Production in Terrestrial Ecosystems

- In terrestrial ecosystems, temperature and moisture affect primary production on a large scale.
- Actual evapotranspiration can represent the contrast between wet and dry climates.
- ***Actual evapotranspiration*** is the *water annually transpired by plants and evaporated from a landscape*. It is related to net primary production.
- On a local scale, a soil nutrient is often the limiting factor in primary production.

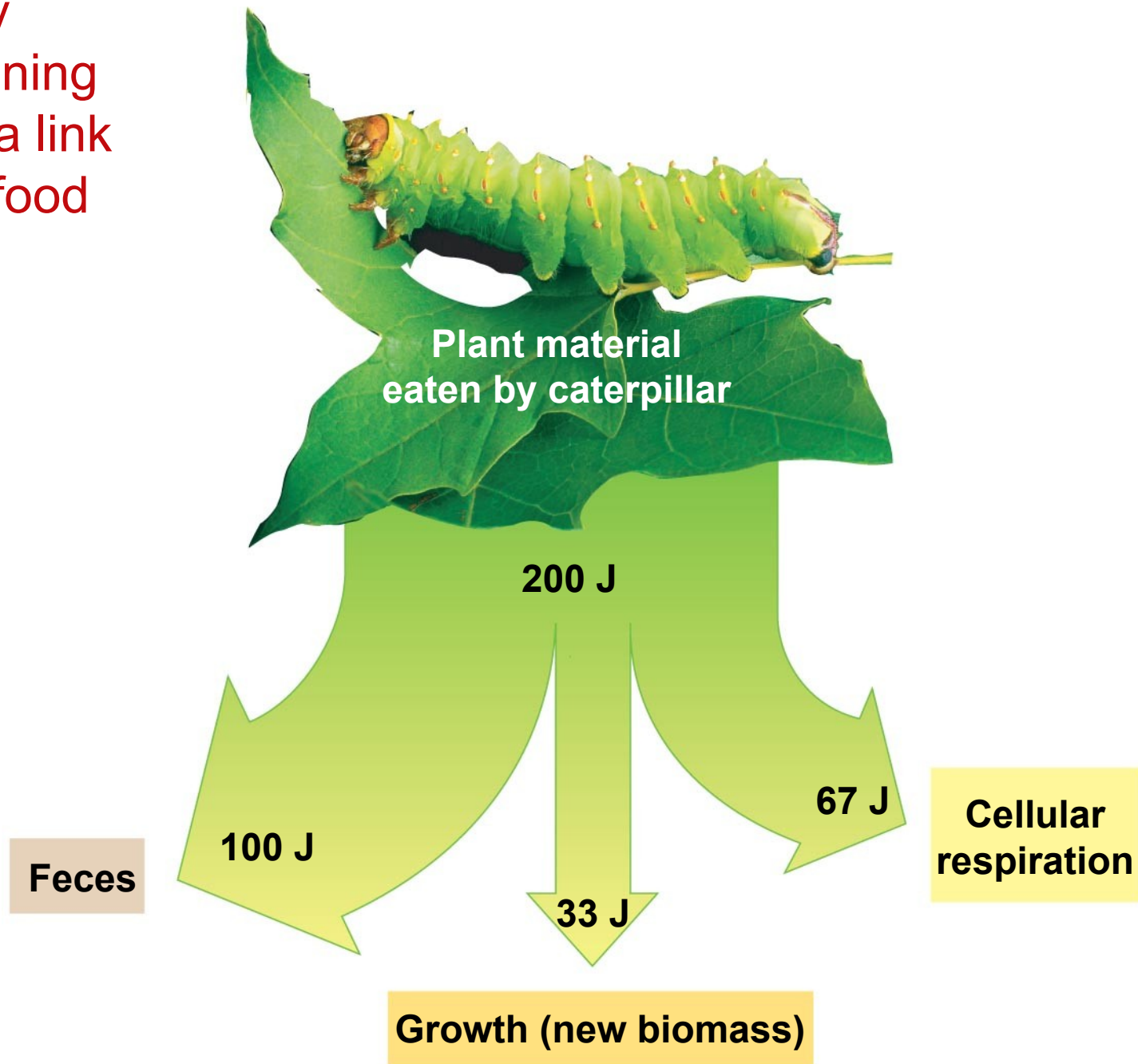
Relationship between **net primary production** and actual **evapotranspiration** in six



Energy transfer between trophic levels is typically only 10% efficient

- **Secondary production** of an ecosystem is the amount of chemical energy in food converted to new biomass during a given period of time.
- When a caterpillar feeds on a leaf, only about one-sixth of the leaf's energy is used for secondary production.
- An organism's **production efficiency** is the fraction of energy stored in food that is not used for respiration.

Energy partitioning within a link of the food chain



Trophic Efficiency and Ecological Pyramids

- **Trophic efficiency** is the percentage of production transferred from one trophic level to the next. *10% Law of Energy Transfer*
- Trophic efficiency is multiplied over the length of a food chain.
- Approximately 0.1% of chemical energy fixed by photosynthesis reaches a tertiary consumer.
- A pyramid of net production represents the loss of energy with each transfer in a food chain.

**Tertiary
consumers**



10 J

**Secondary
consumers**



100 J

**Primary
consumers**



1,000 J

**Primary
producers**

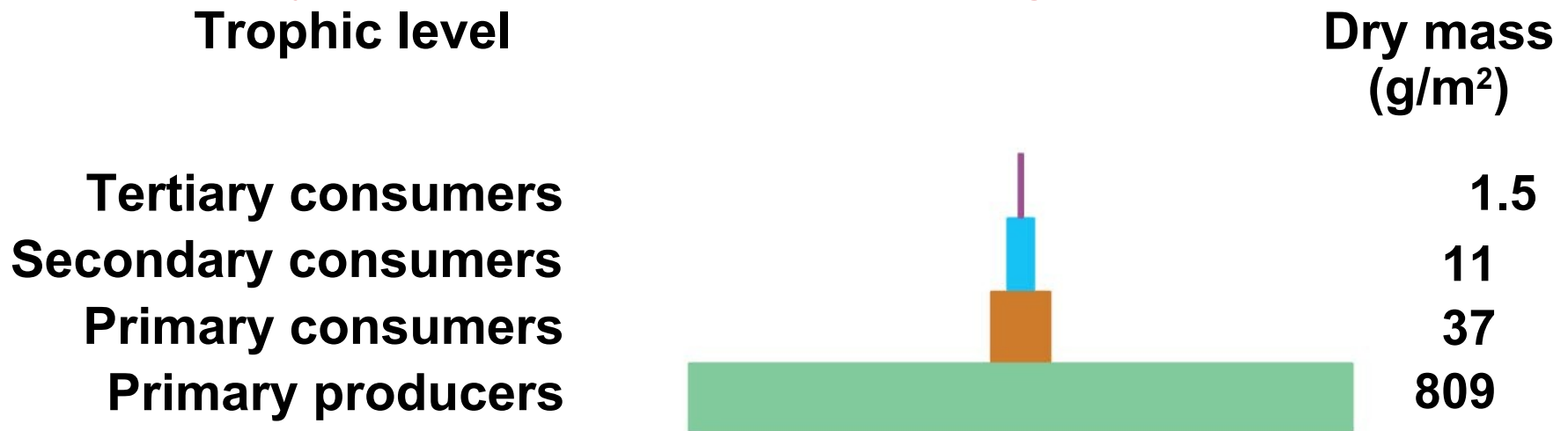


10,000 J

1,000,000 J of sunlight

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- In a biomass pyramid, each tier represents the dry weight of all organisms in one trophic level.
 - Most biomass pyramids show a sharp decrease at successively higher trophic levels..
 - Certain aquatic ecosystems have inverted biomass pyramids: producers (phytoplankton) are consumed so quickly that they are outweighed by primary consumers.
 - **Turnover time** is a ratio of the standing crop biomass to production.

Pyramids of biomass = standing crop:



(a) **Most ecosystems** (data from a Florida bog)



(b) **Some aquatic ecosystems** (data from the English Channel)

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- Dynamics of energy flow in ecosystems have important implications for the human population.
 - Eating meat is a relatively inefficient way of tapping photosynthetic production.
 - Worldwide agriculture could feed many more people if humans ate only plant material.
 - Most terrestrial ecosystems have large standing crops despite the large numbers of herbivores.

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- The **green world hypothesis** proposes several **factors that keep herbivores in check**:
 - Plant defenses
 - Limited availability of essential nutrients
 - Abiotic factors
 - Intraspecific competition
 - Interspecific interactions

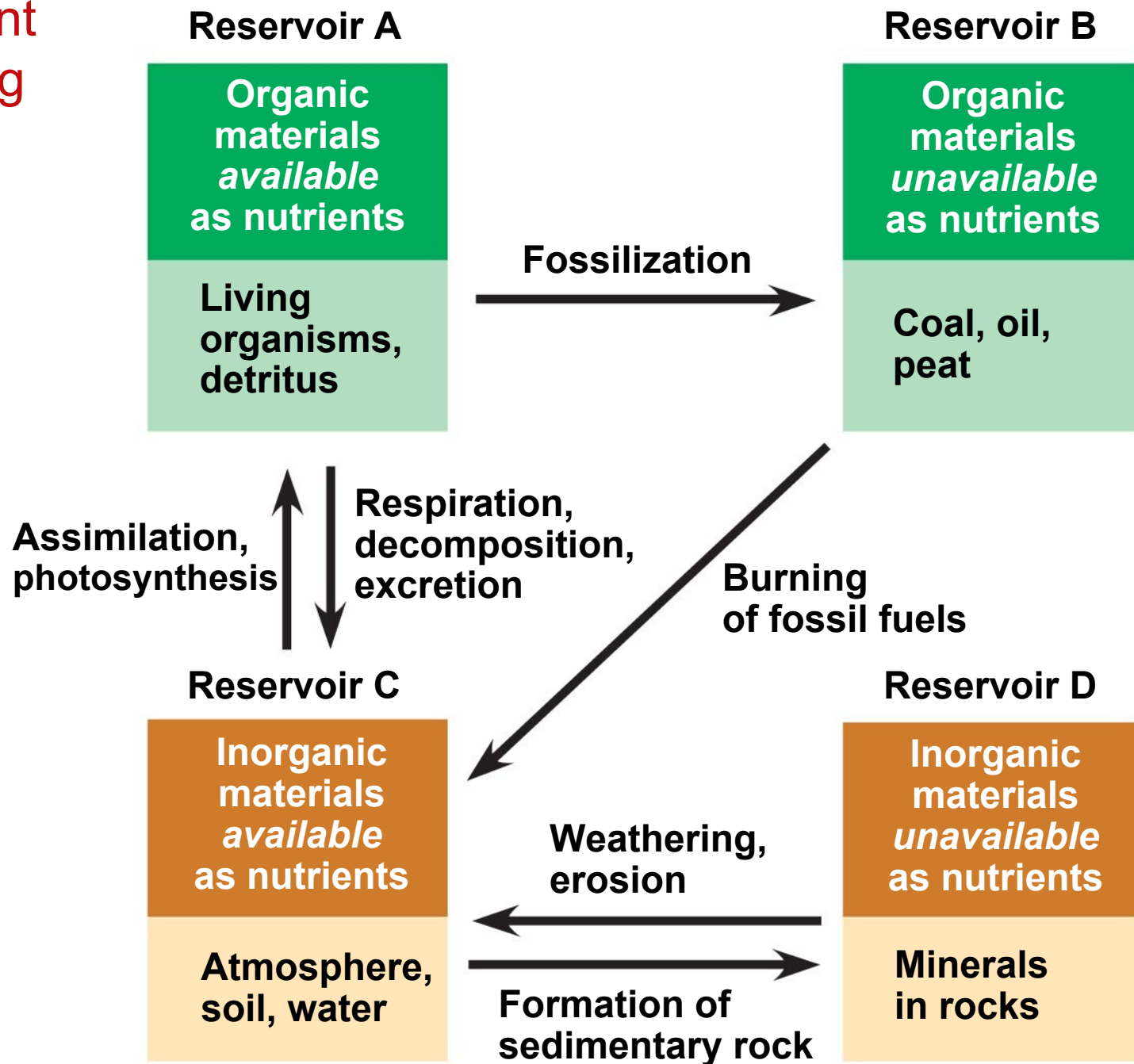
Biological and geochemical processes cycle nutrients between organic and inorganic parts of an ecosystem

- Life depends on recycling chemical elements.
- Nutrient circuits in ecosystems involve biotic and abiotic components and are often called **biogeochemical cycles**.

Biogeochemical Cycles

- Gaseous carbon, oxygen, sulfur, and nitrogen occur in the atmosphere and cycle globally.
- Less mobile elements such as phosphorus, potassium, and calcium cycle on a more local level.
- A model of nutrient cycling includes main reservoirs of elements and processes that transfer elements between reservoirs.
- All elements cycle between organic and inorganic reservoirs.

Nutrient Cycling

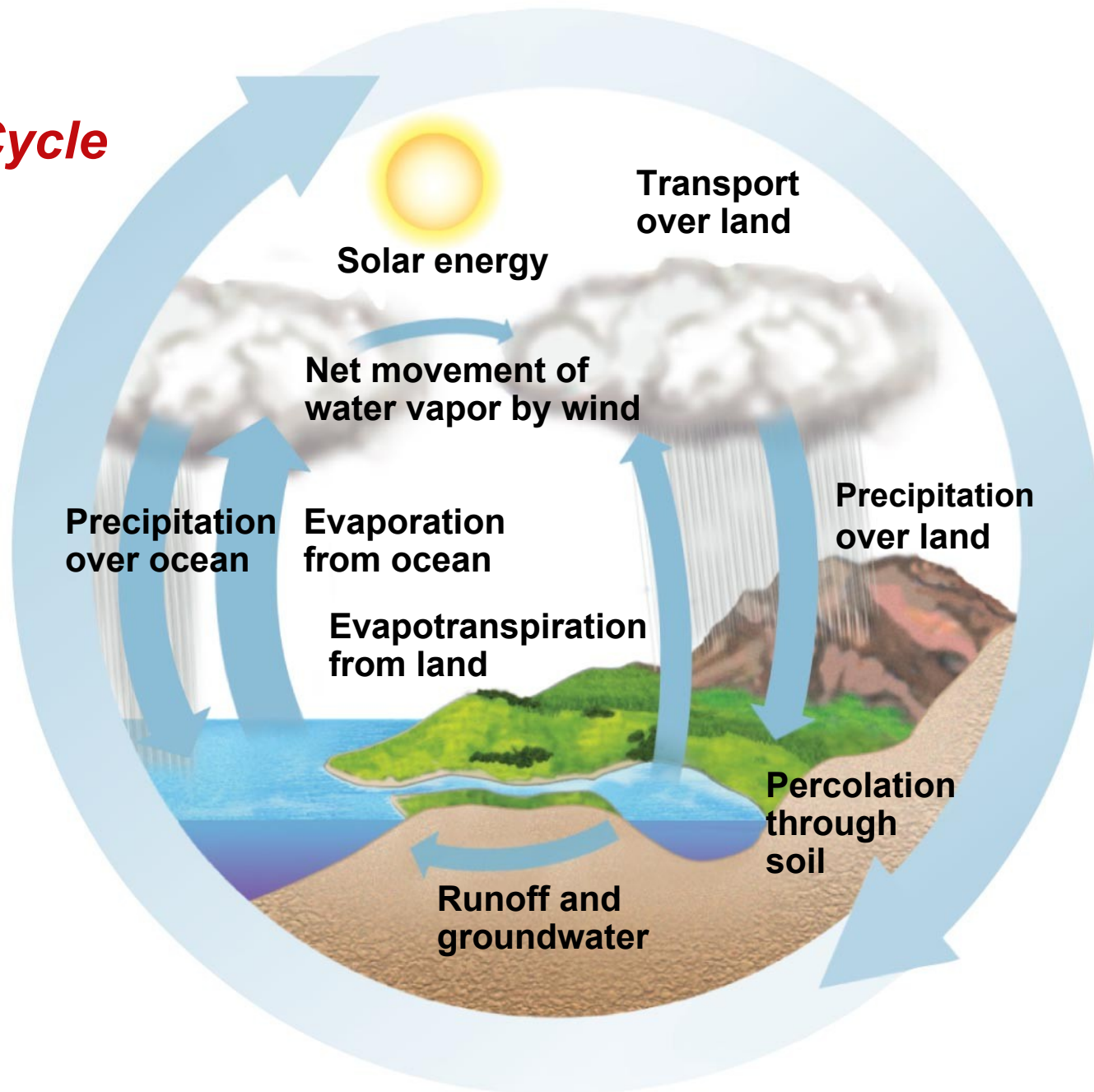


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- In studying **cycling of water, carbon, nitrogen, and phosphorus**, ecologists focus on four factors:
 - Each chemical's **biological importance**
 - **Forms** in which each chemical is **available or used by organisms**
 - Major **reservoirs** for each chemical
 - Key **processes driving movement** of each chemical through its cycle.

The Water Cycle

- Water is essential to all organisms.
- 97% of the biosphere's water is contained in the oceans, 2% is in glaciers and polar ice caps, and 1% is in lakes, rivers, and groundwater.
- Water moves by the processes of evaporation, transpiration, condensation, precipitation, and movement through surface and groundwater.

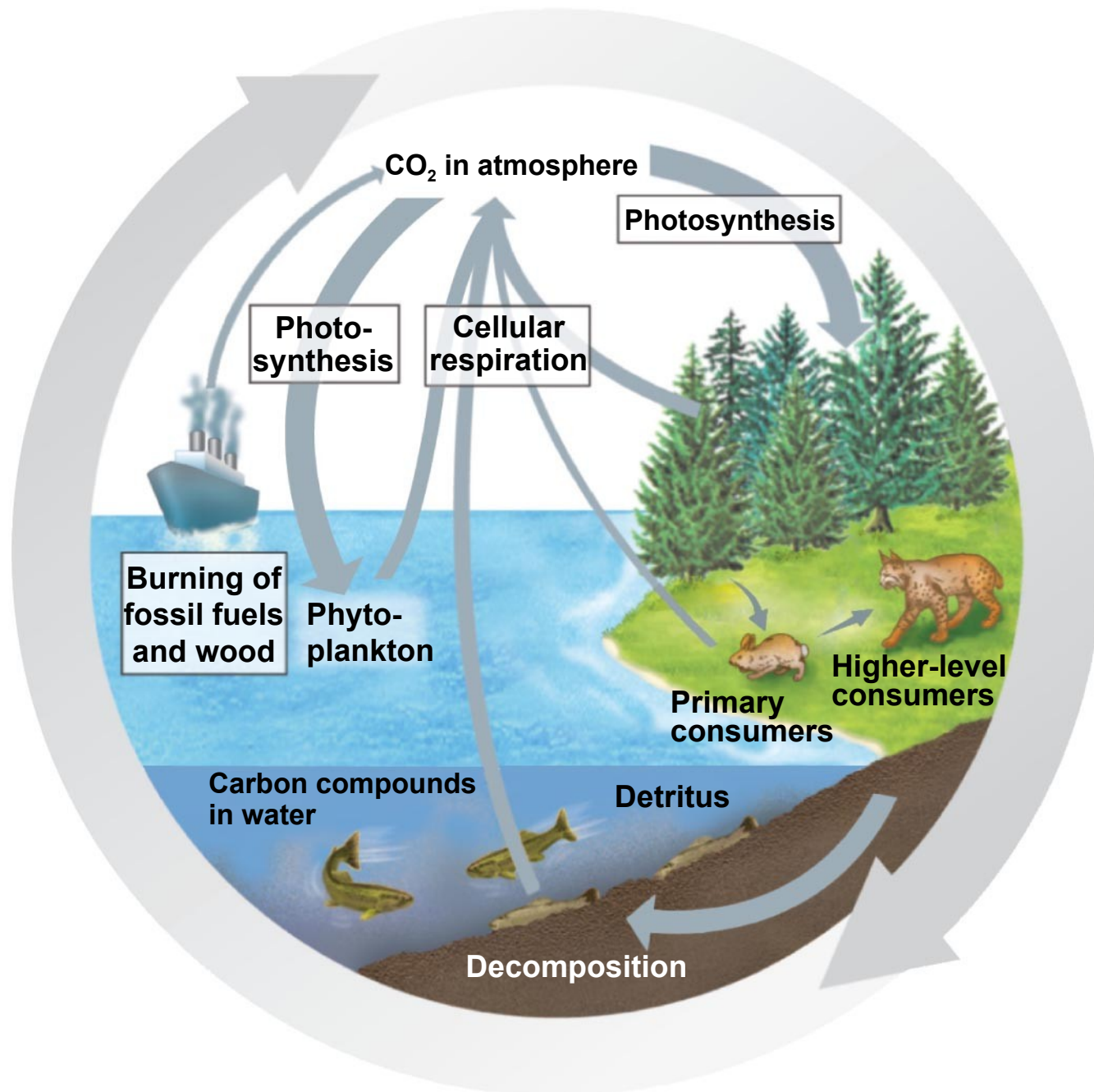
Nutrient Cycles: *Water Cycle*



The Carbon Cycle

- Carbon-based organic molecules are essential to all organisms.
- Carbon reservoirs include fossil fuels, soils and sediments, solutes in oceans, plant and animal biomass, and the atmosphere.
- CO₂ is taken up and released through photosynthesis and respiration; additionally, volcanoes and the burning of fossil fuels contribute CO₂ to the atmosphere.

Nutrient Cycles: *Carbon Cycle*

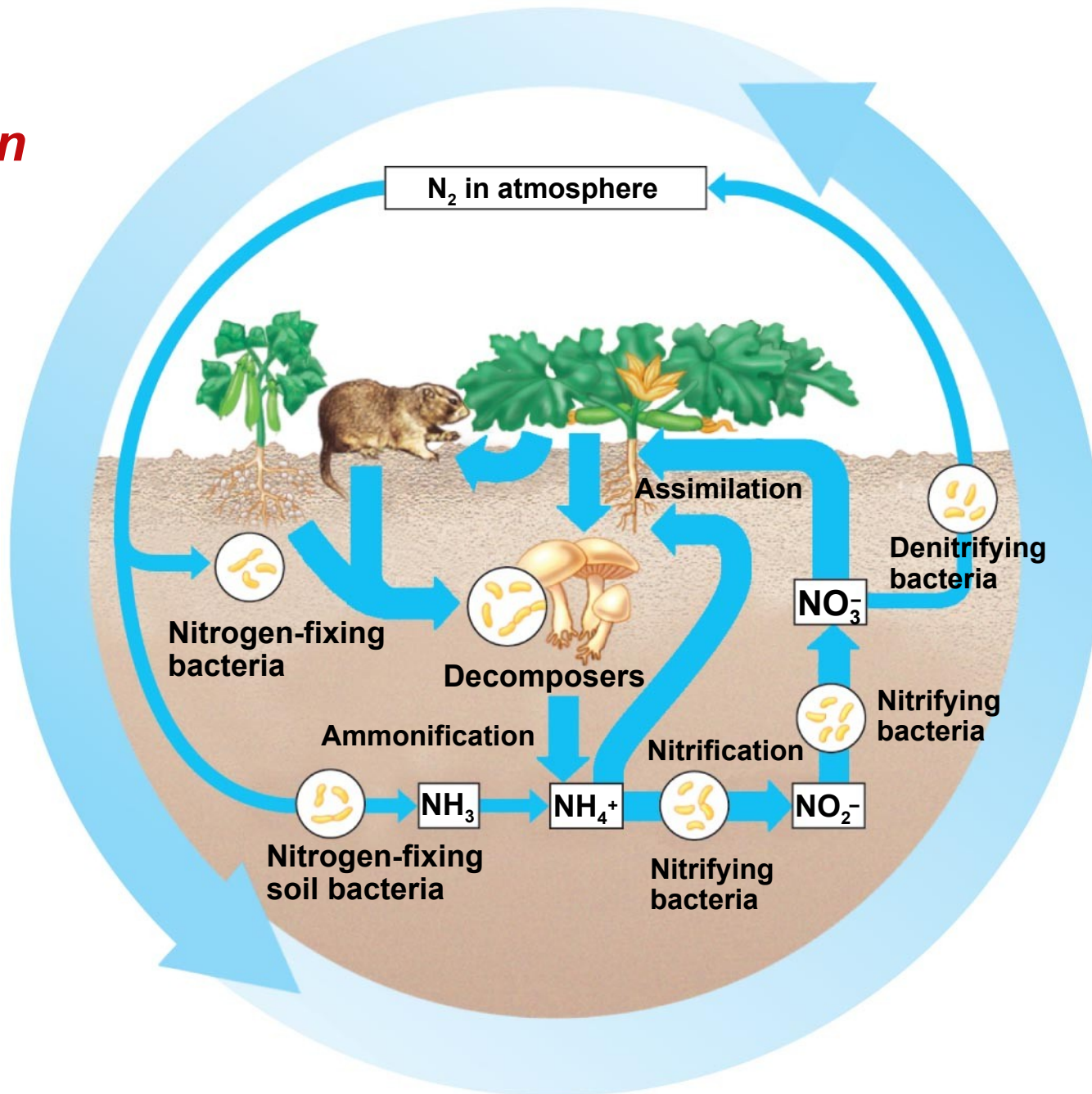


The Terrestrial Nitrogen Cycle

- Nitrogen is a component of amino acids, proteins, and nucleic acids
- The main reservoir of nitrogen is the atmosphere (N_2), though this nitrogen **must be converted to NH_4^+ or NO_3^- for uptake by plants**, via nitrogen fixation by bacteria.
- Organic nitrogen is decomposed to NH_4^+ by ammonification, and NH_4^+ is decomposed to NO_3^- by nitrification.
- Denitrification converts NO_3^- back to N_2

Nutrient Cycles:

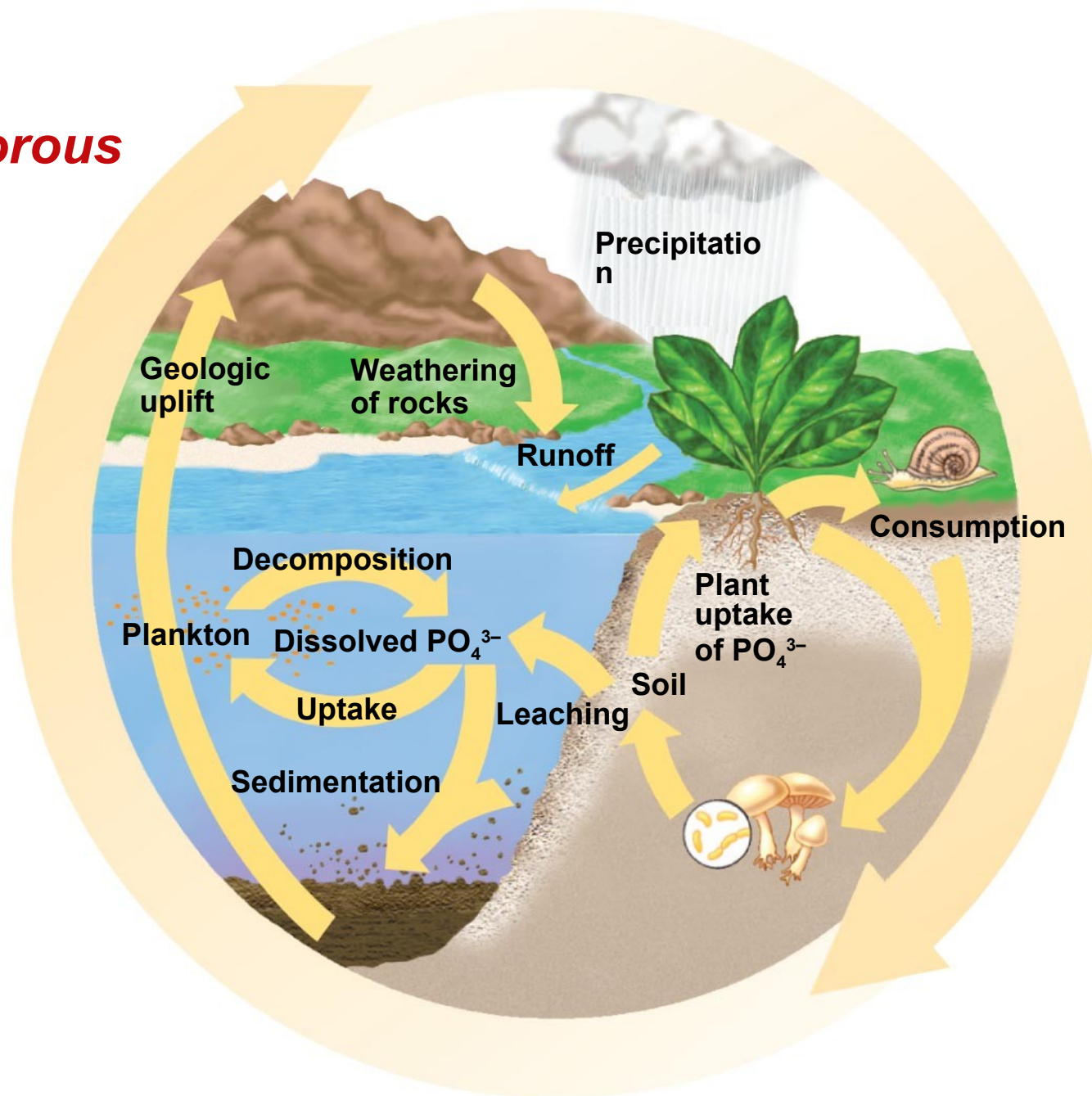
Nitrogen Cycle



The Phosphorus Cycle

- Phosphorus is a major constituent of nucleic acids, phospholipids, and ATP.
- Phosphate (PO_4^{3-}) is the most important inorganic form of phosphorus.
- The largest reservoirs are sedimentary rocks of marine origin, the oceans, and organisms.
- Phosphate binds with soil particles, and movement is often localized.

Nutrient Cycles: *Phosphorous* Cycle

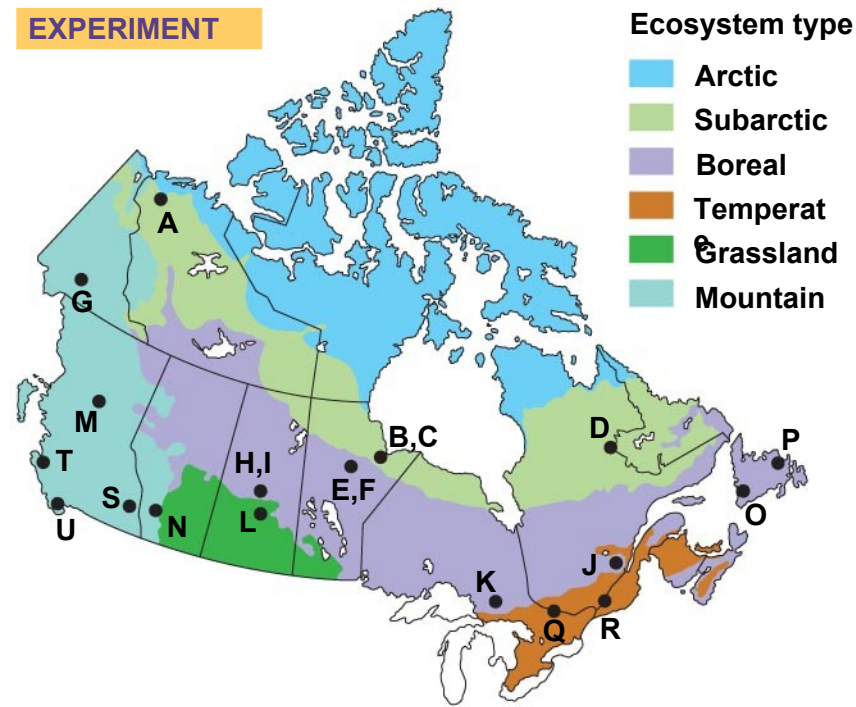


Decomposition and Nutrient Cycling Rates

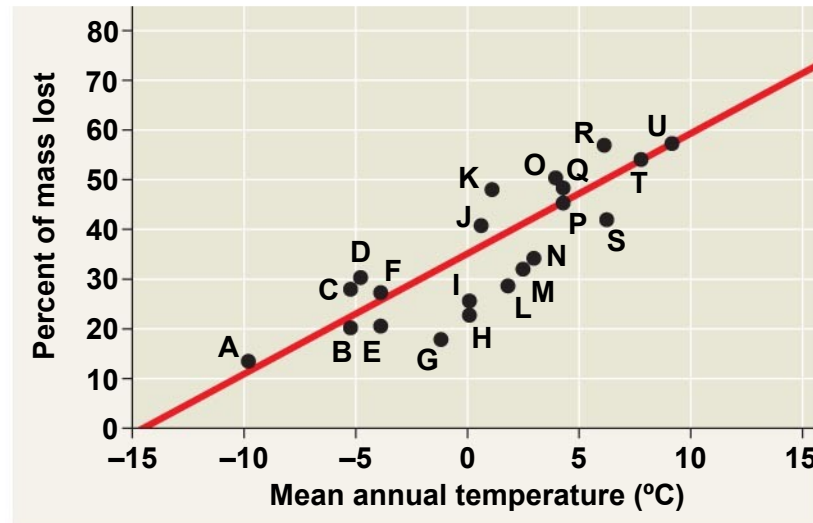
- **Decomposers = detritivores** play a key role in the general pattern of **chemical cycling**.
- Rates at which nutrients cycle in different ecosystems vary greatly, mostly as a result of differing rates of decomposition.
- *The rate of decomposition is controlled by temperature, moisture, and nutrient availability.*
- Rapid decomposition results in relatively low levels of nutrients in the soil.

How does temperature affect litter decomposition in an ecosystem?

EXPERIMENT



RESULTS



Case Study: Nutrient Cycling in the Hubbard Brook Experimental Forest

- **Vegetation strongly regulates nutrient cycling.**
- Research projects monitor ecosystem dynamics over long periods.
- The Hubbard Brook Experimental Forest has been used to study nutrient cycling in a forest ecosystem since 1963.
- The research team constructed a dam on the site to monitor loss of water and minerals.

Nutrient Cycling

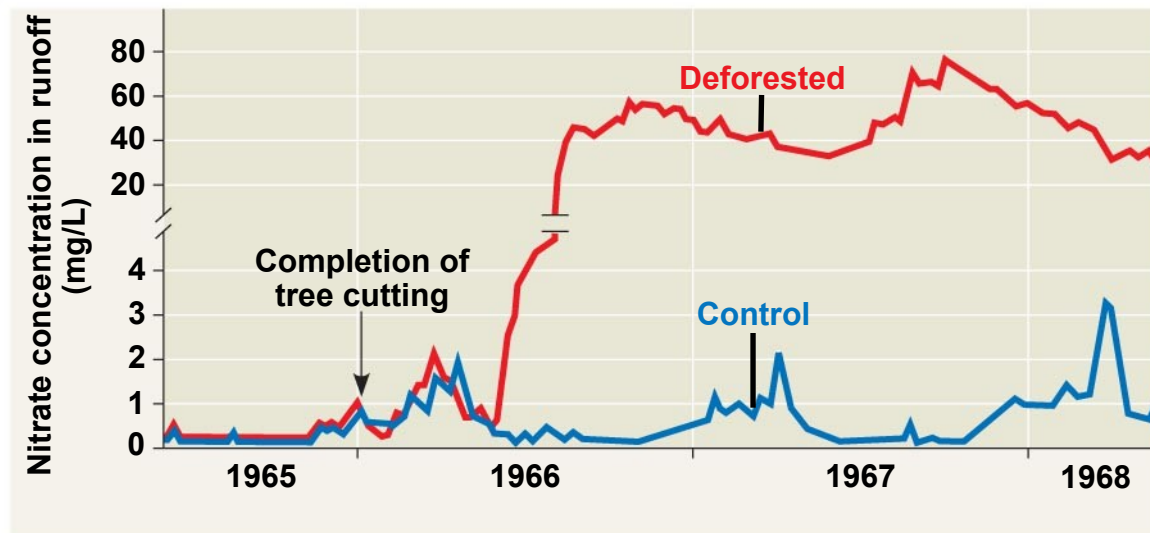
in the
Hubbard Brook
Experimental
Forest: an
example of
long-term
ecological
research



(a) Concrete dam and weir



(b) Clear-cut watershed



(c) Nitrogen in runoff from watersheds

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- In one experiment, the trees in one valley were cut down, and the valley was sprayed with herbicides. Net losses of water and minerals were studied and found to be greater than in an undisturbed area.
 - These results showed how human activity can affect ecosystems.
 - *As the human population has grown, our activities have disrupted the trophic structure, energy flow, and chemical cycling of many ecosystems.*
 - In addition to transporting nutrients from one location to another, humans have added new materials, some of them toxins, to ecosystems.

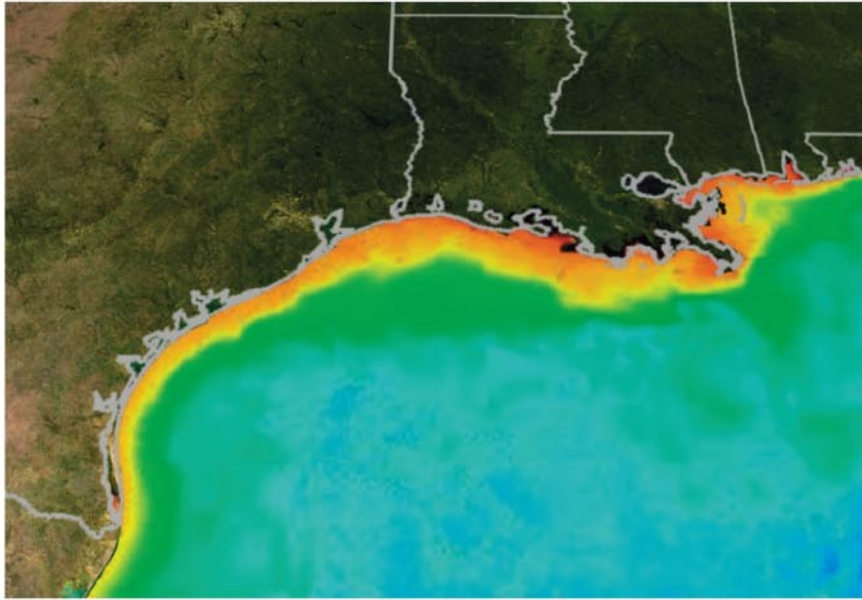
Agriculture and Nitrogen Cycling

- The quality of soil varies with the amount of organic material it contains.
- Agriculture removes from ecosystems nutrients that would ordinarily be cycled back into the soil.
- Nitrogen is the main nutrient lost through agriculture; thus, agriculture greatly affects the nitrogen cycle.
- Industrially produced fertilizer is typically used to replace lost nitrogen, but effects on an ecosystem can be harmful.

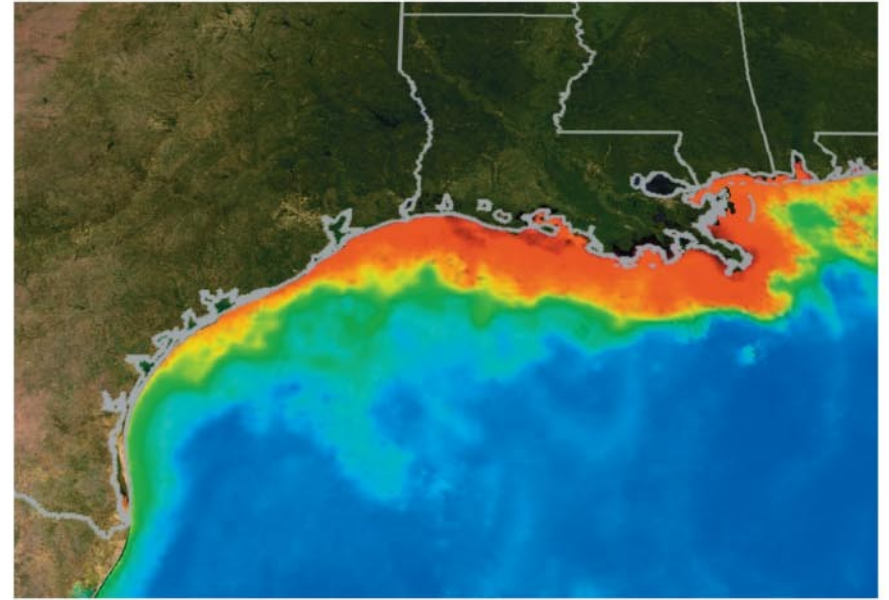
Contamination of Aquatic Ecosystems

- **Critical load** for a nutrient is the amount that plants can absorb without damaging the ecosystem.
- **When excess nutrients** are added to an ecosystem, the **critical load is exceeded**.
- Remaining **nutrients can contaminate** groundwater as well as **freshwater and marine ecosystems**.
- Sewage **runoff causes** cultural **eutrophication**, excessive algal growth that can greatly harm freshwater ecosystems.

The dead zone arising from nitrogen pollution in the Mississippi basin



Winter

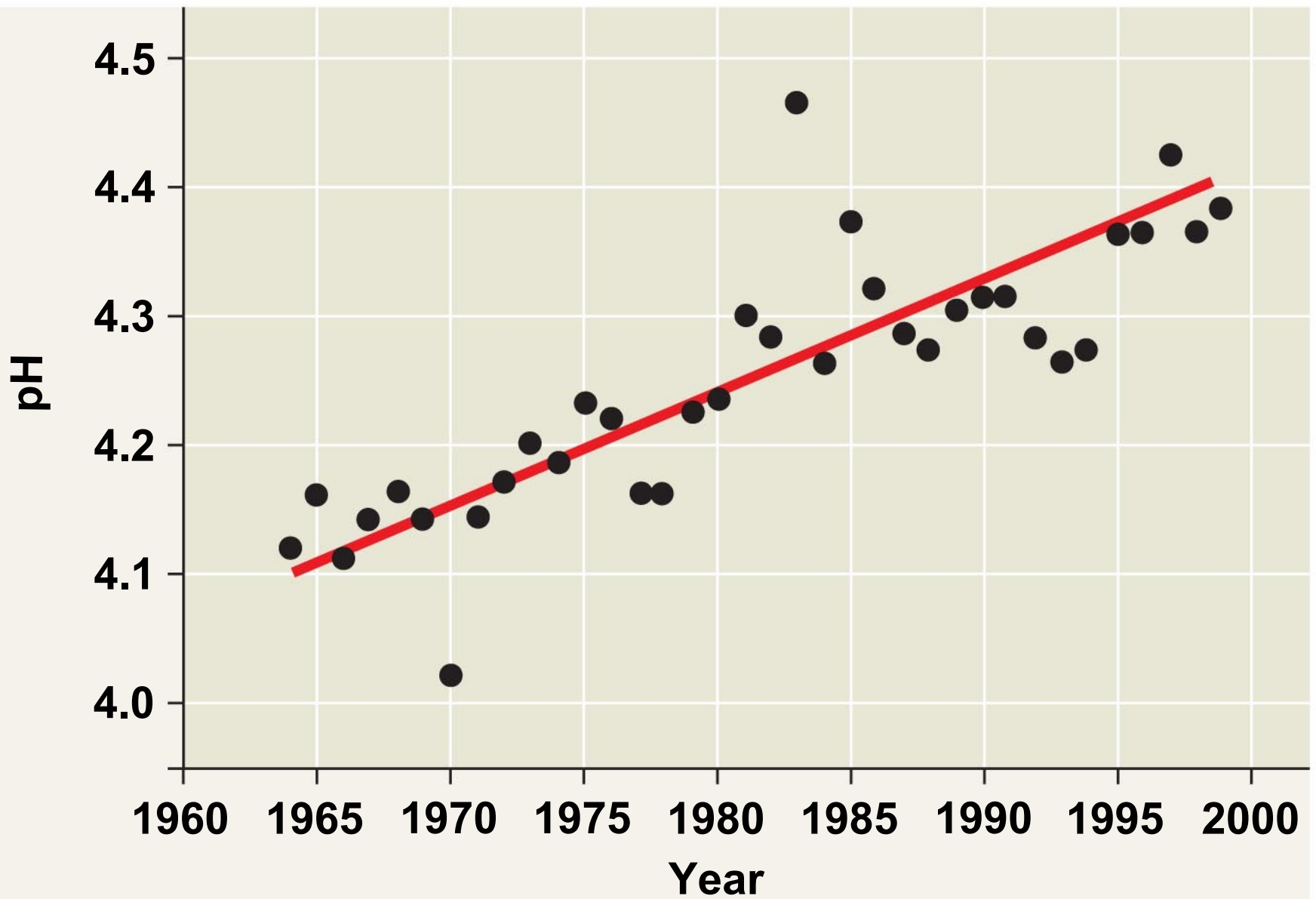


Summer

Acid Precipitation

- *Combustion of fossil fuels is the main cause of acid precipitation.*
- North American and European ecosystems downwind from industrial regions have been damaged by rain and snow containing **nitric and sulfuric acid**.
- *Acid precipitation changes soil pH and causes leaching of calcium and other nutrients.*
- Environmental regulations and new technologies have allowed many developed countries to reduce sulfur dioxide emissions.

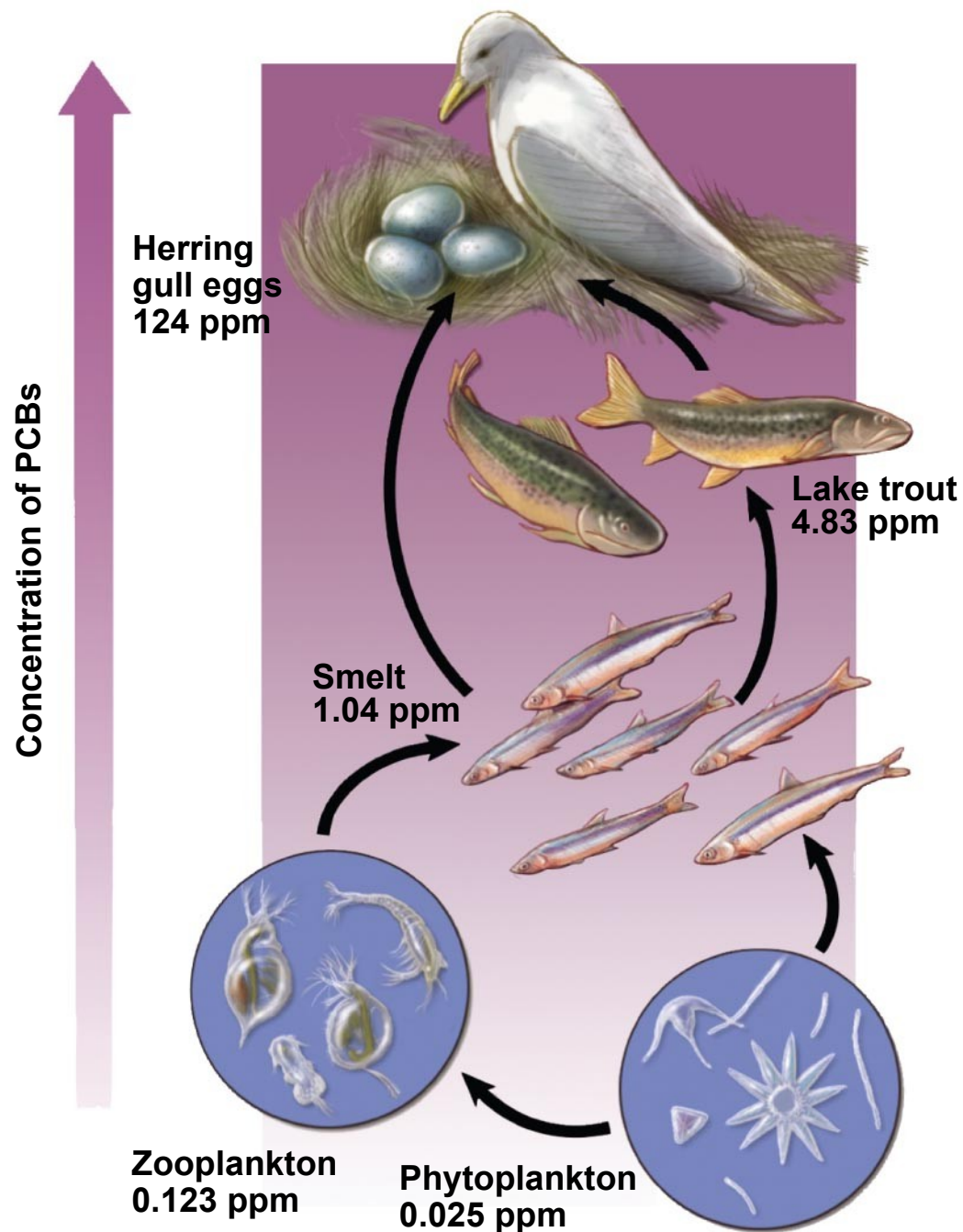
Changes in the pH of precipitation at Hubbard Brook



Toxins in the Environment

- Humans release many toxic chemicals, including synthetics previously unknown to nature. In some cases, harmful substances persist for long periods in an ecosystem.
- One reason **toxins** are harmful is that they become **more concentrated in successive trophic levels**.
- **Biological magnification** concentrates toxins at higher trophic levels.
- PCBs and many pesticides such as DDT are subject to biological magnification in ecosystems.

Biological Magnification of PCBs in a Great Lakes food web



Greenhouse Gases and Global Warming

- One pressing problem caused by human activities is the rising level of atmospheric carbon dioxide.
- Due to the burning of fossil fuels and other human activities, the concentration of atmospheric CO₂ has been steadily increasing.

The Greenhouse Effect and Climate

- CO₂, water vapor, and other greenhouse gases reflect infrared radiation back toward Earth; this is the **greenhouse effect**.
- This effect is important for keeping Earth's surface at a habitable temperature.
- **Increased levels** of atmospheric CO₂ are magnifying the greenhouse effect, which could cause **global warming** and **climatic change**.

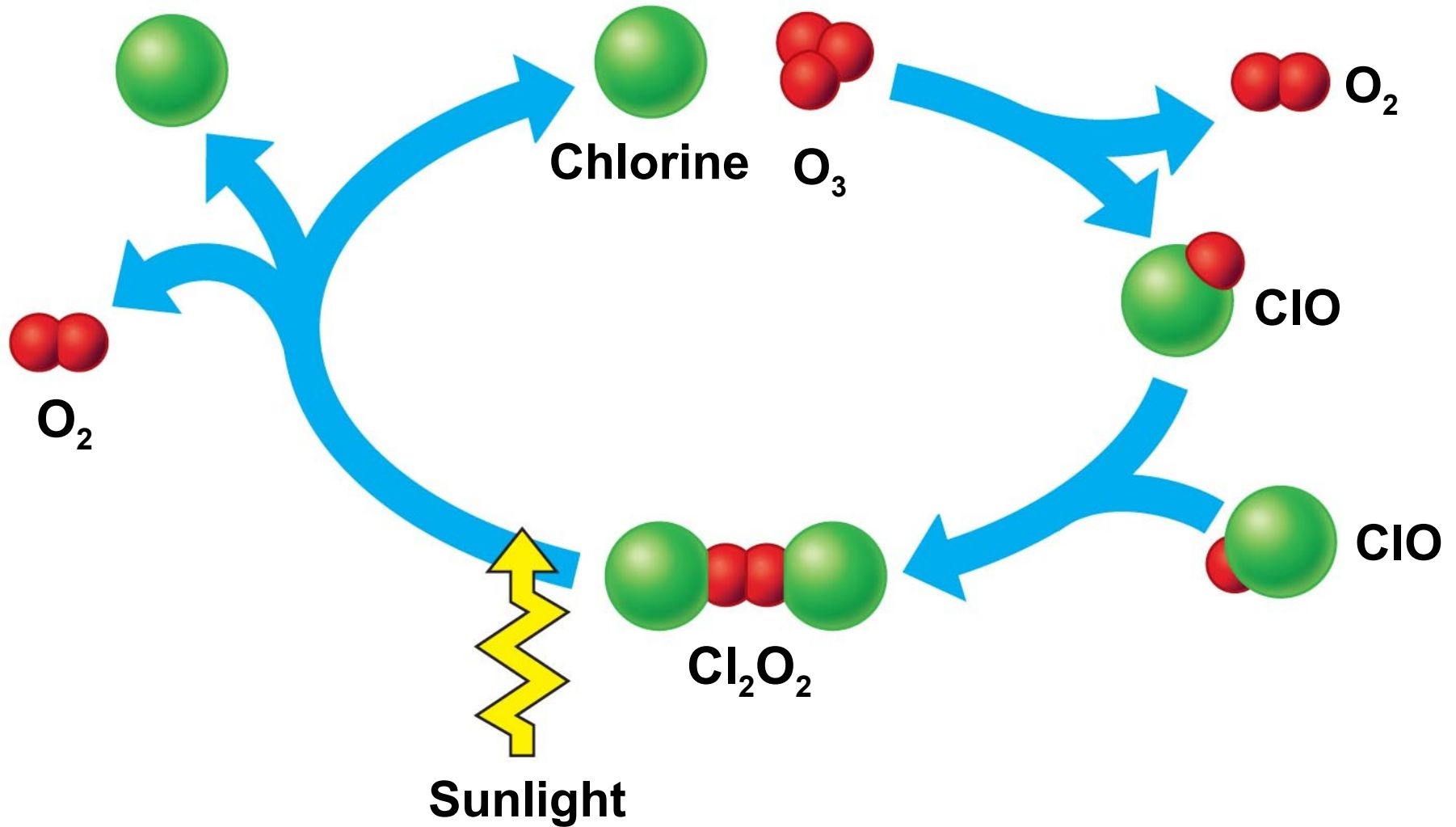
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- Increasing concentration of atmospheric CO₂ is linked to increasing global temperature. Northern coniferous forests and tundra show the strongest effects of global warming.
 - A warming trend would also affect the geographic distribution of precipitation.
 - Global warming can be slowed by reducing energy needs and converting to renewable sources of energy.
 - Stabilizing CO₂ emissions will require an international effort.

Depletion of Atmospheric Ozone

- Life on Earth is protected from **damaging effects of UV radiation** by a protective layer of **ozone molecules in the atmosphere**.
- Satellite studies suggest that the ozone layer has been gradually thinning since 1975.
- **Destruction of atmospheric ozone** probably results from chlorine-releasing pollutants such as **CFCs** (chloroflorocarbons) produced by human activity.

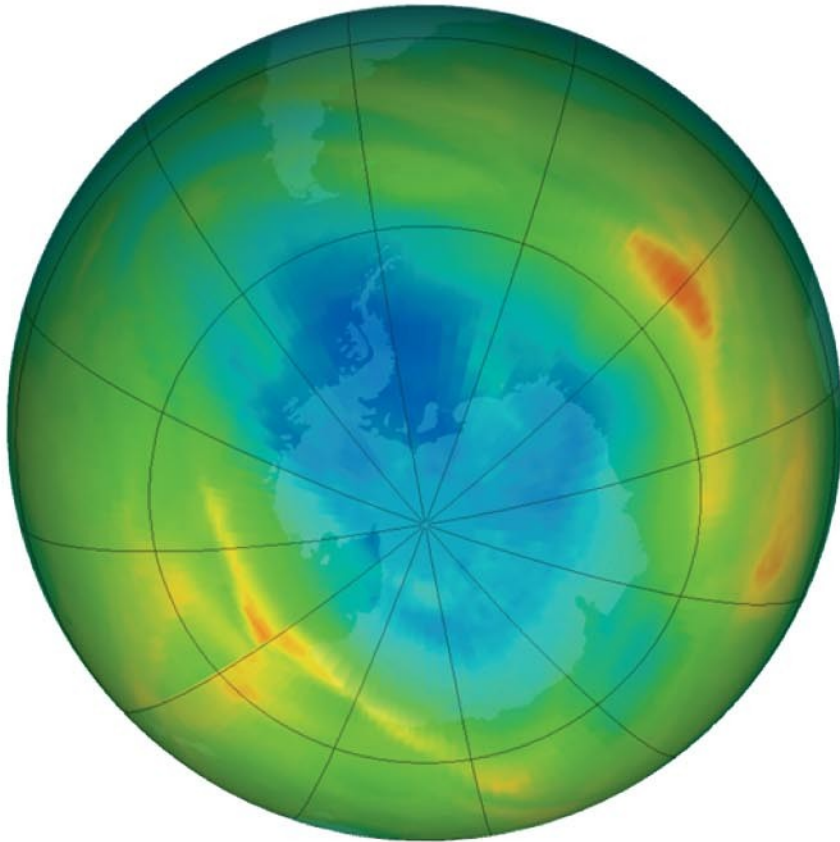
How free chlorine in the atmosphere destroys ozone

Chlorine atom

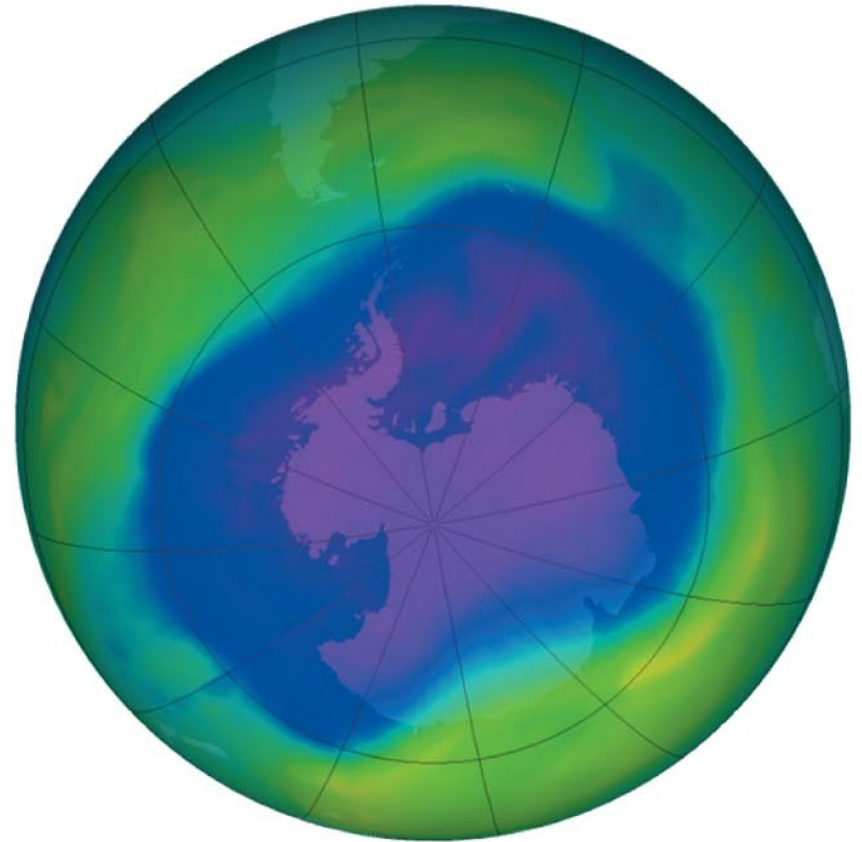


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- Scientists first described an “ozone hole” over Antarctica in 1985; it has increased in size as ozone depletion has increased.
 - Ozone depletion causes DNA damage in plants and poorer phytoplankton growth.
 - An international agreement signed in 1987 has resulted in a decrease in ozone depletion.

Erosion of Earth's ozone shield

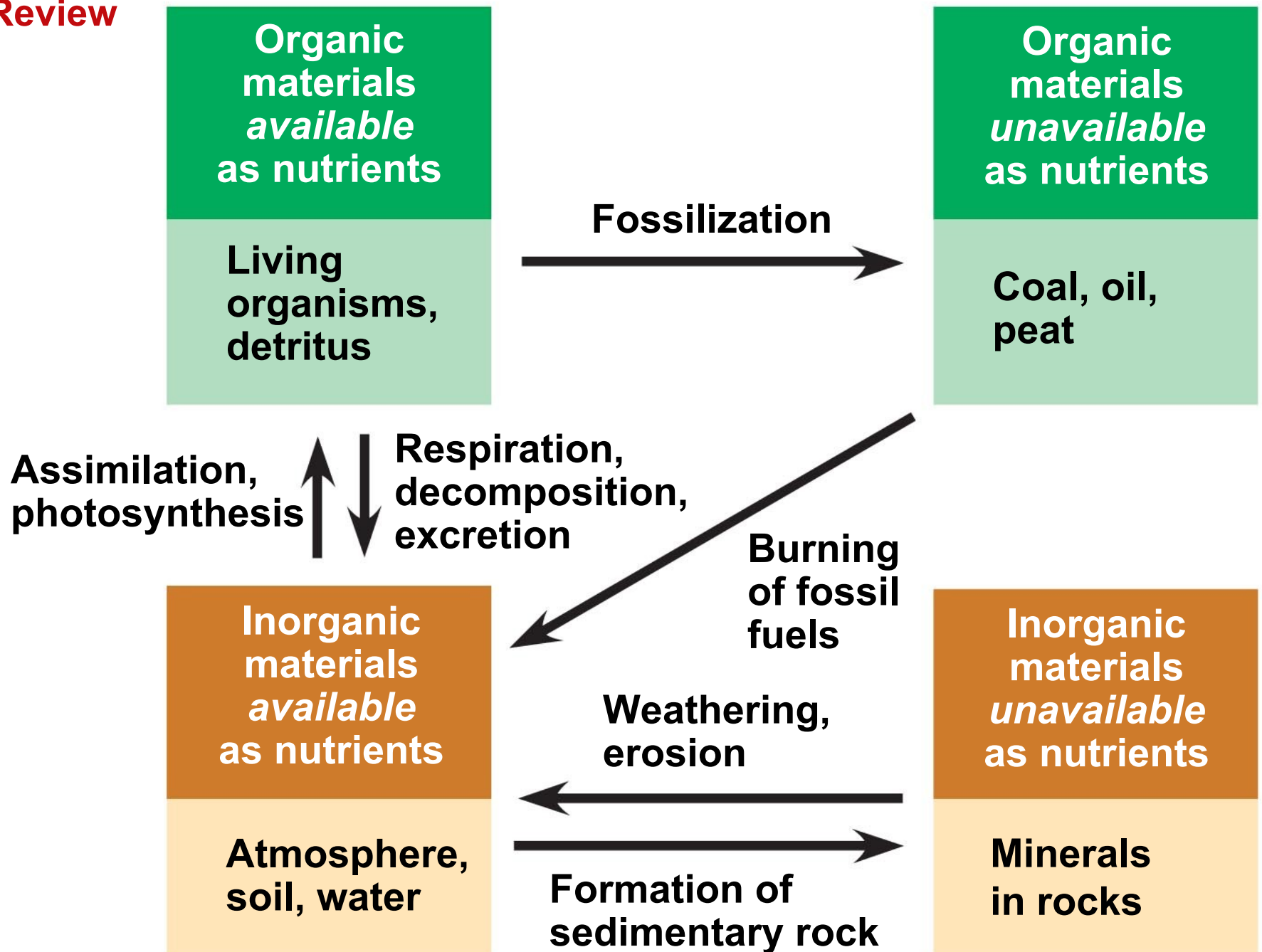


(a) September 1979



(b) September 2006

Review



You should now be able to:

1. Explain how the first and second laws of thermodynamics apply to ecosystems.
2. Define and compare gross primary production, net primary production, and standing crop.
3. Explain why energy flows but nutrients cycle within an ecosystem.
4. Explain what factors may limit primary production in aquatic ecosystems.

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5. Distinguish between the following pairs of terms: primary and secondary production, production efficiency and trophic efficiency.
 6. Explain why worldwide agriculture could feed more people if all humans consumed only plant material.
 7. Describe the four nutrient reservoirs and the processes that transfer the elements between reservoirs.

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8. Explain why toxic compounds usually have the greatest effect on top-level carnivores.
 9. Describe the causes and consequences of ozone depletion.